

CHAPTER 6

Work Design and Organizational Performance—Work Measurement¹

6.1. Introduction

"Managers at all levels and in all organizations are striving to achieve results through people."² To do this all sorts of control systems have been designed to measure and enhance organizational performance. Unfortunately, many of the popular control systems monitor only a small part of total system performance. Recently, however, great strides have been made toward the design of effective and efficient control systems. Industrial engineers are leading the way.

In the first edition, this chapter focused on work measurement and design at the workplace. That was an accurate portrayal of industrial engineering involvement at the time but was microscopic or myopic in that it did not measure organizational performance. It did measure individual efficiency. In the second edition we modernized that a bit by talking more about true "productivity," as explained below. Thus, the area is becoming more system or total system performance oriented.

Now, the third edition recognizes that this trend is continuing and perhaps

¹The authors are deeply indebted to Dr. D. Scott Sink and Ms. Leva K. Swim for their guidance and help with this chapter.

²D. Scott Sink, "Organizational System Performance: Is Productivity a Critical Component?" *Proceedings 1983 Annual Industrial Engineering Conference*, Louisville, Ky., May 1983.

accelerating. In fact, many industrial engineering curricula are dropping the more traditional areas of workplace measurement and improvement. We feel, however, that it is important to keep this in perspective. Thus, this chapter will briefly cover traditional areas and move on to the more company-wide techniques.

Perhaps Figure 6.1 does the best job of placing all this in perspective. Organizational productivity is simply a measurement of an organization's ability to turn inputs into outputs. Equation (6.1) states this:

$$\text{productivity} = \frac{\text{output}}{\text{input}} \quad (6.1)$$

Outputs and inputs can be partial or composite. For example, units per person-hour is a measure of labor productivity only and is, therefore, partial. Assuming that we desire quality outputs which meet the consumer demand (effectiveness), we can see that productivity is affected by efficiency, effectiveness, and quality, as shown in Figure 6.1. Productivity, together with innovation and quality of working life, determine the total organizational performance, which is usually measured by profitability in the free-enterprise system.

Now, where does work design and measurement (timing of individual operations) fit? Referring to Figure 6.1, we see that *work measurement is the standard by which we measure efficiency*. Similarly, *work design helps determine efficiency*. Work measurement and design can be distinguished from total organizational performance through considerations of *focus* and *scope*.

Total organizational system performance has a much broader *focus* in that it addresses more performance criteria than just efficiency (i.e., quality, quality of working life, effectiveness, innovation, profitability, productivity). Work measurement/design in the workplace has a narrow focus in that it addresses efficiency, primarily. In regard to *scope*, total organizational system performance is concerned with the collection of individuals organized to accomplish a purpose. Work measure-

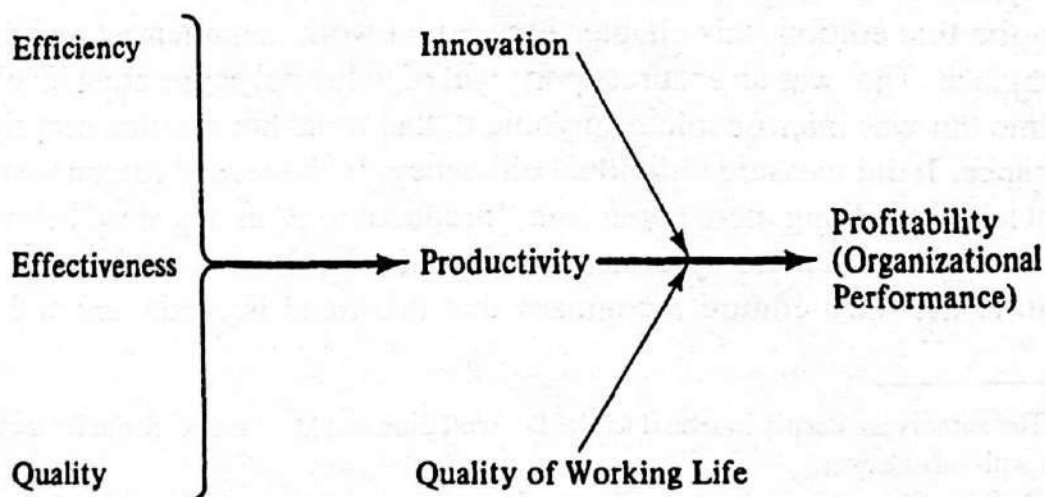


Fig. 6.1. Total organization performance. [Adapted from D.S. Sink, *Productivity Management: Planning Measurement, Evaluation, Control, and Improvement* (New York: John Wiley & Sons, Inc., 1984)].

ment/design concerns itself with the individual person and the person's interaction with his or her environment.

This chapter examines the measurement of organizational performance and the design of individual workstations. It does that by first discussing work simplification (probably the birthplace of industrial engineering) and then work measurement. Then the chapter discusses organizational performance with an emphasis on productivity measurement.

Some of you may never do work methods simplification or work measurement, as many industrial engineers are branching out into different areas. However, you should understand that industrial engineering started here and that more I.E.s are employed in this area today than in any other. It is sometimes very helpful to know your discipline's birthplace.

The following example is presented to demonstrate the need for the topics of this chapter.

Example 6.1

You have been contacted by the president of Gadgets, Inc., in Corvallis, Oregon, and asked to design and lay out the manufacturing processes for production of their new "super-gadget." Furthermore, he wants you to stay on and manage the plant once it is completed.

Since Gadgets, Inc., has done a great deal of research on processing methods, you find that there is little work needed there. Your goal is to produce 10,000 units per year. You must gear up to produce that, *but* to be able to determine how much equipment is necessary, you must know how long each operation takes. Since Gadgets, Inc., does not have that kind of information, you hire a time study analyst to do work measurement. Furthermore, since the equipment is expensive, you want the most efficient methods possible. The analyst must also design the most efficient work procedures for the workers to follow.

With the results of the time study analyst's work, you have the information necessary for determining manpower needs. Again, you find you must know how many super-gadgets each person can produce. You go back to the previously taken time studies to determine your labor needs.

Of course, the people must be trained. Who is going to train them, and what methods should they use? You decide to let your foremen do the training, *and* you find that the time study analyst specified the proper method for each job. The training is then simplified.

Because you are human relations oriented, you want to pay your people on the basis of their work quantity and quality. One way of doing this is to have an *incentive system*³ in which the people are paid according to how much they do. The previously taken time studies provide you with the time information you need for this incentive system.

Next, it is necessary to start production (construction and layout are complete). One of the first things you must do is schedule the orders for super-gadgets (each order has a unique requirement or design change) throughout the shop. To do this, you must figure how long a production order takes for each operation and the total time necessary in the plant. Since your time studies provide you with this information, you can do the scheduling. Production now begins.

³Incentive systems are discussed in Chapter 9.

As you sit back in your chair, you reflect, "How can I know if the plant is performing as it should?" Once again, the time studies give you a bench mark against which you can compare performances since you know how long the operations should take.

Finally, you need to decide how much to charge for super-gadgets. There are many considerations, for example, the market supply, but one consideration is the cost of production. Since the plant is just starting, you do not know the cost of production. The time studies tell you what the labor cost *should* be. You then add on material and overhead cost and you are ready.

Or, at least you thought you were. The president recently attended a productivity improvement seminar; so he tells you to design a productivity measurement model for the company. Using the material of this chapter, you find you can, indeed, measure productivity. In fact, you feel the need to inform the president that he also needs to worry about total organizational performance (hopefully, profitability) and that productivity is only a part of total performance.

Although the above example is grossly oversimplified, it does point out *some* of the uses of work simplification and measurement. Those uses were in:

- (1) Determining equipment requirements.
- (2) Determining labor requirements.
- (3) Designing training methods.
- (4) Designing scheduling procedures.
- (5) Designing incentive systems.
- (6) Gauging work performance.
- (7) Estimating cost.

This is not an exhaustive list by any means, but it should be sufficient to demonstrate the importance of work simplification and measurement.

We have demonstrated that the correct design of work methods and timing of these methods is very important to an organization, but how can these be done? For obvious reasons, work simplification should be performed on a task before that task is timed. Therefore, it is logical to examine some procedures for work simplification first.

6.2. Methods Improvement

Over the evolutionary sequence of industrial engineering, work methods improvement has been called many things, including work simplification, methods improvement, and work design. Whatever it is called (we chose the term *work methods improvement*) a definition might be: *A systematic approach to finding easier and better ways to accomplish a task.*

The basic objective is to avoid waste of any kind (time, human effort, materials, capital, etc.) or, as often quoted

Work smart, not hard.⁴

You have been making attempts at work methods improvement all your life whenever you try to find better ways to cut grass, wash cars, travel to work, and so on, but have you used a systematic approach? In some cases, probably yes, but in others, no. To do work methods improvement correctly, a systematic approach is necessary. We will discuss some systematic approaches in this section. A five-step procedure might be:

- (1) *Selecting a Job*—One must select a job to study. This is often done because the job is the most expensive or obviously in need of such a study.
- (2) *Getting and Recording the Facts*—The present method of doing the job must be recorded along with other facts such as the frequency of the job.
- (3) *Questioning Every Detail*—The present method must be questioned. Some elements may be deleted; most can be changed.
- (4) *Developing and Testing a Better Method*—From step 3, better efforts may be obvious. These ideas should be developed and tried.
- (5) *Installing and Maintaining Improvements*—For the study to be worthwhile an improved procedure must be installed and maintained.

Charts are a convenient way of applying parts of the foregoing procedure, especially in steps 2 and 3. The operations process chart already covered is one. Some of the other charts will be discussed next in their approximate order of application.

6.2.1 Flow Process Charts

A flow process chart is a chart of all the activities involved in a process. It is similar to an operations process chart, except that more detail is shown by including transportations and delays as well as operations, inspections, and storages. The symbols used are explained in Chapter 4. As in operations process charts, the objective is to list every detail and to examine each detail closely. The following questions should be asked about each activity:

- (1) Is this activity necessary, or can it be eliminated?
- (2) Can this activity be combined with another or others?
- (3) Is this the proper sequence of activities, or should the sequence be changed?
- (4) Can this activity be improved?
- (5) Is this the proper person to be doing this activity?

The construction is similar to that of an operations process chart in that flow is from the top to the bottom. Only one item can be charted per form, however,

⁴Stephan Konz, *Work Design: Industrial Ergonomics* (Columbus, Ohio: Grid Publishing, Inc., 1983), p. 21.

and that item can be either a person or an object. A blank flow process chart is shown in Figure 6.2.⁵ An example of its use is demonstrated below.

Example 6.2

Mr. Cafe, a school food cafeteria supervisor, has asked you to prepare a flow process chart for the preparation of spaghetti. You have decided to make a chart of the noodles going through the cafeteria and see how the process can be improved.

The noodles start on the storage shelf. They are picked up and carried to the cooker, which is a large vat in which the noodles are cooked. Next they are placed in a large bowl, which is carried to a sink where they are washed. Then the bowl is carried to a work area where the noodles are placed in six serving pans. The meat and sauce (a separate flow process chart covers their production) are added, and the pans are carried to a heating unit where they are kept warm until ready for serving.

You prepare a flow process chart (shown in Figure 6.3). After consultation with Mr. Cafe and the operators and examining the chart carefully, you decide that by adding a water tap adjacent to the cooker (it already has a drain) you could wash the noodles right in the cooker, add the meat and sauce directly, and keep them warm in the cooker. This saves several transportations. The revised procedure is shown in Figure 6.4.

The advantages of the revised procedure over the existing one are obvious and numerous. No heavy lifting is required, only three transportations are required—a saving of four (57%)—and the same number of operations and storages are required.

In the existing procedure, steps 4 and 8 are called *transportations*, but they could have been called *operations*. Either term would have worked very well since the term is not important *in cases like this*. Many times an activity can be called several things; usually all are shaded on the chart, or only the most predominant ones are shown, as was done here.

6.2.2 Left-Hand-Right-Hand Charts

Left-hand-right-hand charts are useful in analyzing the work performed by one person at one specific workstation. As the name implies, the chart follows the motion of the left and right hands of one operator. Consequently, it is applicable only when one operator is performing repetitive work—usually at one station. An example is given below.

⁵This is a simplified form in that much more information usually is presented. It is slightly different from the form used in Chapter 4; but either form or another can be used, depending on the preference of the user.

Process Begins Charter Present _____ Proposed _____	Man _____ or Material _____ Ends Date
Steps 1. 2. 3. 4. 5. :	Symbol
Notes	

Fig. 6.2. Flow process chart.

Process: Prepare Spaghetti Begins: In Storage Charter: Ind. Eng. Present <u>X</u> Proposed _____	Man ____ or Material <u>X</u> Noodles Ends: Ready to Serve Date: 7/24
Steps 1. Noodles on Shelf 2. Carried to Cooker 3. Cooked 4. Placed in Bowl 5. Carried to Sink 6. Washed 7. Bowl Carried to Work Area 8. Placed in Serving Pans 9. Meat and Sauce Added to Pans 10. Pans Carried to Heater 11. Warmed 12. Carried to Serving Line	Symbol ▽ ◇ ○ ◇ ◇ ○ ◇ ◇ ○ ◇ ▽ ◇
Notes Wash in Cooker? Very Heavy Very Heavy Six Pans-Full Repeated Six Times Repeated Six Times Repeated Six Times	

Fig. 6.3. Flow process chart—present method.

Process: Prepare Spaghetti Begins: In Storage Charter: Ind. Eng Present ☐ Proposed ☒		Man ☐ or Material ☒ Noodles Ends: Ready to Serve Date: 7/26
Steps	Symbol	Notes
1. Noodles on Shelf	▽	
2. Carried to Cooker	◇	
3. Cooked	○	
4. Washed in Cooker	○	Add a Tap Adjacent to Cooker
5. Meat and Sauce Added	○	
6. Warmed	▽	
7. Placed in Serving Pans	◇	Repeat Six Times
8. Carried to Serving Line	◇	Repeat Six Times

Fig. 6.4. Flow process chart—proposed method.

Example 6.3

In this example a pot handle is formed by plastic injection molding. This process requires a sophisticated die, and consequently, much time is devoted to the design of the die. Regardless of the amount of time spent on the design, a small burr is left on the part after it is removed from the die. This burr is ground in a small grinder; then it is polished by a buffing wheel to eliminate any grinding marks. The process used is presented below. Plastics, Inc., has asked you to improve the process and you have decided to use a left-hand-right-hand chart. First, you ask for a drawing of a typical burr. That is presented in Figure 6.5. Then you ask for a sketch of the workplace and a description of the process. The workplace sketch is shown in Figure 6.6.

All pot handles are placed on the gravity chute by a material handler. The chute is inclined so that parts are always available to the operator. The operator reaches to the chute with her left hand and picks up three pot handles. In the center of the work station she holds the handles with her left hand. Her right hand picks one pot handle out of the left hand, grinds the pot handle (takes about 2 seconds), and drops it in a cardboard box on the floor marked for completed units. The right hand does all three parts and the left hand reaches for three more pot handles. This is continued approximately 56 times until the cardboard box on the floor is full. The operator stops, shoves the box over to the area marked full boxes, and

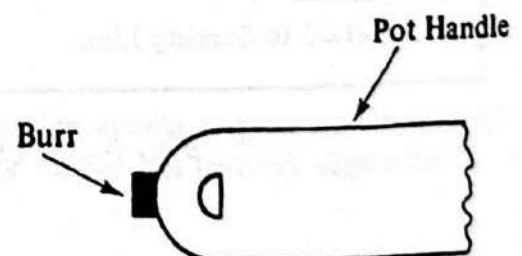


Fig. 6.5. Typical burr.

The symbols are similar to those of a flow process chart, for example:



Operation: A hand is doing some productive work



Transport: A hand is moving from one place to another



Delay: A hand is not accomplishing anything but is waiting



Hold: A hand is holding an object.

Example 6.3 (Continued)

You construct a left-hand-right-hand chart for the operation. That chart is shown in Figure 6.8. You notice that the right hand is delayed a large portion of the time while it is waiting for the left hand to obtain the parts. Also, the left hand holds the parts while the right hand does the work. (Holding is really not doing any productive work.) Productive work then really only occupies a small percentage of the total cycle.

After working with the operator and the foreman and after trying several new methods, you come up with a proposed procedure. (You also consider the principles of motion economy that will be discussed in Section 6.2.4.) That proposed procedure is shown in Figure 6.9 with a sketch of the new work plan. You time the two procedures and find that you save 37% of the time formerly used, not counting the substantial material handling savings.

Now there are two chutes feeding the operator from the left and the right. There is also a work shelf added to the grinding table. The work shelf has a drop hole in it directly in front of the wheel. The operator reaches simultaneously left and right and slides two parts on the work shelf (one per hand) to the two grinding wheels. After the grinding occurs, the operator slides the parts to the middle where they drop through to a conveyor underneath. The conveyor carries the handles to the polishing operation. The cycle is repeated continuously because boxes never have to be moved.

6.2.3 Other Charts

This discussion only skims the surface of charting for work simplification. There are many other charts that have been used in designing work for many different types of jobs. Some of the other popular charts are described briefly below.

Flow Diagram. A flow diagram is essentially a flow process chart drawn to show the layout of a facility. Only the symbols and the step numbers are shown. Usually, the objective is to look for spatial relationships.

Operation Grinding Burrs Name Bear Brian Date 7/30

Existing X Proposed _____

Left-hand Activity	Symbol	Symbol	Right-hand Activity
1. Reach to Chute	◇	□	1. Wait
2. Pick up 3 Parts	○	□	2. Wait
3. Move to Center	◇	□	3. Wait
4. Hold Parts	▽	○	4. Position Part and Grind
5. Hold Parts	▽	◇	5. Move to Right and Drop in Box
6. Hold Parts	▽	◇	6. Move Back to Center
7. Hold Parts	▽	○	7. Repeat 4
8. Hold Parts	▽	◇	8. Repeat 5
9. Hold Parts	▽	◇	9. Repeat 6
10. Hold Parts	▽	○	10. Repeat 4
11. Hold Parts	▽	◇	11. Repeat 5
Repeat Steps 1 through 8 56 times.			
9. Walk to Box	□	□	9. Same as left hand
10. Slide Full Box Out	◇	◇	10. Same as left hand
11. Move Empty Box In	◇	◇	11. Same as left hand
12. Walk Back	□	□	12. Same as left hand

Fig. 6.8. Left-hand-right-hand chart—existing procedure.

Operation Grinding Burrs Name Ara Smith Date 7/30

Existing Proposed X

Left-hand Activity			Right-hand Activity
1. Reach to and Preposition Handle	◇	◇	1. Reach to and Preposition Handle
2. Slide to Grinder	◇	◇	2. Slide to Grinder
3. Grind	○	○	3. Grind
4. Slide to Center and Drop	◇	◇	4. Slide to Center and Drop

Fig. 6.9. Left-hand-right-hand chart—proposed procedure.

Multiple Activity Chart. A multiple activity chart is similar to a left-hand-right-hand chart in that the simultaneous activities of two or more objects or persons are shown. A chart simultaneously showing the activities of a worker and those of one or more machines he is operating is an example. The simultaneous activity of several workers on an assembly line is another. The objective is to obtain good balance—minimizing the total cycle time and subsequently the waiting time of any of the components.

Work Distribution Chart. A work distribution chart is a listing of all activities or responsibilities of every person in a department or group. It is similar to a multiple activity chart except that the activities are not necessarily simultaneously performed. The objective is to obtain proper balance of assignments and to be certain that the proper level of employment is performing the function (e.g., a department head should not deliver mail or answer telephone calls).

Operation Process Chart (already discussed)—The objective is to eliminate unnecessary operations, inspections, or storages and to obtain the proper sequence of them.

Gantt Chart. A Gantt chart is a horizontal time bar chart that shows relative timing of various activities. It is not often used in work simplification but may be used in scheduling. The objective is usually to perform all activities in the least amount of total time—often subject to some resource availability constraints.

6.2.4 Principles of Motion Economy

The *principles of motion economy* is a list of thoughts or concepts that have proven helpful in designing efficient work methods. There have been many lists generated by different authors; but the lists always carry the same essential concepts.

Many industrial engineers use motion economy in conjunction with left-hand-right-hand charts, since the list is most useful for single workstations. There are other uses, however, and one of them is to use the principles of motion economy as a list of commandments in designing and performing work. This last is perhaps the most rewarding but least utilized approach.

The following list of motion economy principles is taken from Mundel.⁶

A. Elimination:

1. Eliminate all possible jobs, steps, or motions. (This applies to body, leg, arm, hand, or eye.)
2. Eliminate irregularities in a job so as to facilitate automaticity. Provide fixed places for things.

⁶Reproduced, with permission, from Marvin E. Mundel, *Motion and Time Study: Improving Productivity*, 5th ed. (Englewood Cliffs, N.J.: Prentice Hall, 1978), pp. 173–174.

3. Eliminate the use of the hand as a holding device.
 4. Eliminate awkward or abnormal motions.
 5. Eliminate the use of muscles to maintain a fixed position.
 6. Eliminate muscular force by using power tools, power feeds, etc.
 7. Eliminate the overcoming of momentum.
 8. Eliminate danger.
 9. Eliminate idle time unless needed for rest.
- B. Combination:**
1. Replace with one continuous curved motion short motions which are connected with sudden changes in direction.
 2. With fixed machine cycles, make a maximum of work internal to the machine cycle.
 3. Combine tools.
 4. Combine controls.
 5. Combine motions.
- C. Rearrangement:**
1. Distribute the work evenly between the two hands. A simultaneous symmetrical motion pattern is most effective. (This frequently involves working on two parts at the same time.) With crew work, distribute the work evenly among members of the crew.
 2. Shift work from the hands to the eyes.
 3. Arrange for a straightforward order of work.
- D. Simplification:**
1. Use the smallest muscle group capable of doing the work, providing for intermittent use of muscle groups as needed.
 2. Reduce eye travel and the number of fixations.
 3. Keep work in the normal work area.
 4. Shorten motions.
 5. Adapt handles, levers, pedals, buttons, etc., to human dimensions and musculatures.
 6. Use momentum to build up energy in place of the intense application of muscular force.
 7. Use the simplest possible combination of therbligs. (A *therblig* is an elementary activity such as reach, grasp, pick up.)
 8. Reduce the complexity of each therblig, particularly the "terminal" therbligs.

6.2.5 Human Engineering

Human engineering (or ergonomics) may be defined as the design of the human-machine interface so that workers and machines may function more effectively and efficiently as integrated systems. Much of what has been covered in this chapter can be called human engineering, but there is much more to human engineering than work simplification and principles of motion economy.

Formerly, machines were designed with an emphasis on their capabilities for production but with little or no emphasis on the human element. Now, with human engineering, the machine may be lowered, tilted, or completely redesigned so that the worker and the machine may function better as a complete system. Much work in human engineering is done in design and location of dials for maximum effectiveness and efficiency. The design of a cockpit for a large modern jet is an excellent example of human engineering.

Human engineering is also useful when applied to day-to-day operations. For example, at an individual workstation it is important that all work be located close to and in front of the worker, as implied by the principles of motion economy. Using human engineering, we can design the workstation for the "average" person. Human engineers have calculated the reach of the average worker for both easy and strained moves. The workstation can be designed so that as many moves as possible are within easy reach. Figure 6.10 is an example of such measurements.

Another example of human engineering is the design of tools for more effective use. Pliers and scissors with curved handles are results of human engineering studies in which it was found that correctly designed curved handles are easier for human beings to use.

6.3. Work Measurement

The objective of any work measurement system is to determine the time it should take an average, trained person to perform a task if he or she were doing that over an 8-hour day under usual working conditions and working at a normal pace. This time is called *standard time*. Historically, there have been two basic approaches to defining standard time. We shall call these two the bottom-up and top-down approaches. The *bottom-up approach* starts with a basic measurement of time, adjusts for operator pace, and then allows for fatigue, personal needs, and delays. The *top-down approach* is used in many labor contracts, and it normally defines standard time as that time under which a qualified employee working under usual conditions *can* make an incentive pay (specified) percent above base pay. Whichever definition is used, the approach taken to calculate standard time is *almost* always the bottom-up approach. Consequently, we shall explore that approach. First, some definitions are in order.

Normal Time. The time required for an average, trained operator to perform a task under usual working conditions and working at a normal pace. (It does not include allowances for personal needs and delays that would be necessary if the task were done all 8 hours.)

Normal Pace. The pace of an average, trained, and conscientious operator working over an 8-hour day.

Actual Time. The observed time required for an operator to perform a task.

Normal work area based approximately on average man's dimensions.

Physical data for average man.

Weight = 155 lbs.

Height = 69 in.

Length of arm = 30 in.

Length of forearm = 18 in.

Length of hand = 7 in.

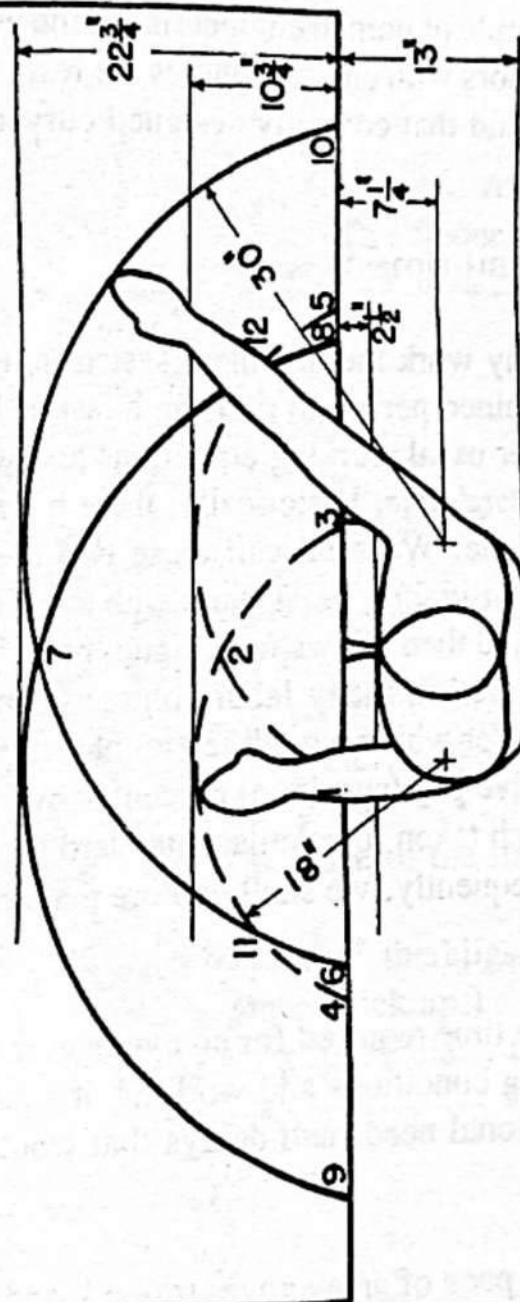


Fig. 6.10. Human work areas.

Allowances. The amount of time added to the *normal time* to provide for personal needs, unavoidable delays, and fatigue.

We can now examine some of the possible methods of determining standard times. Following are the methods to be discussed:

- (1) Direct time study.
- (2) Time study standard data.
- (3) Predetermined time systems.
- (4) Predetermined time systems standard data.
- (5) Work sampling.

6.3.1 Direct Time Study

In this context, direct time study is a work measurement technique in which a physical measurement is made of the actual time required to do a task by using a watch or some other timing device. This measured time is then modified by considering the operator's pace, and finally, allowances are added.

The form used to record task times varies greatly among companies. A general rule of thumb is to present as much information as might be required to duplicate the operation from the form alone. Therefore, the form should include a good description of the task involved, a sketch of the workplace, perhaps a left-hand-right-hand chart of the operation, and a clear description of the elemental breakdown.

Figure 6.11 shows a typical watch used in direct time studies and Figure 6.12 shows a typical time study board. A time study board is a device for holding the forms so that the analyst can make a recording while standing or even moving.

Figure 6.13 shows a simplified sample time study sheet for a continuous

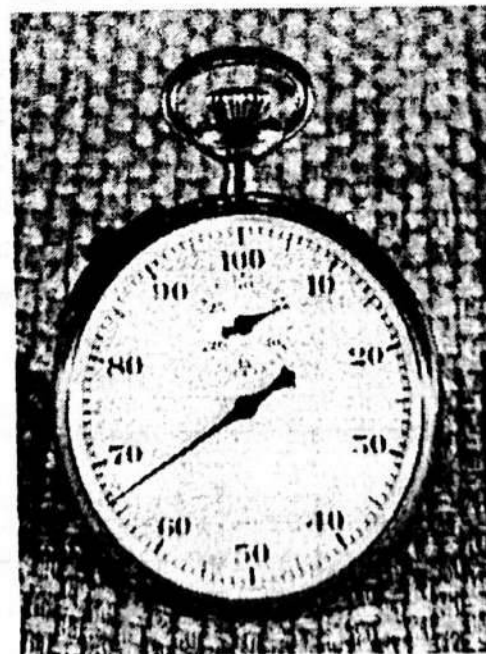


Fig. 6.11. Time study watch.

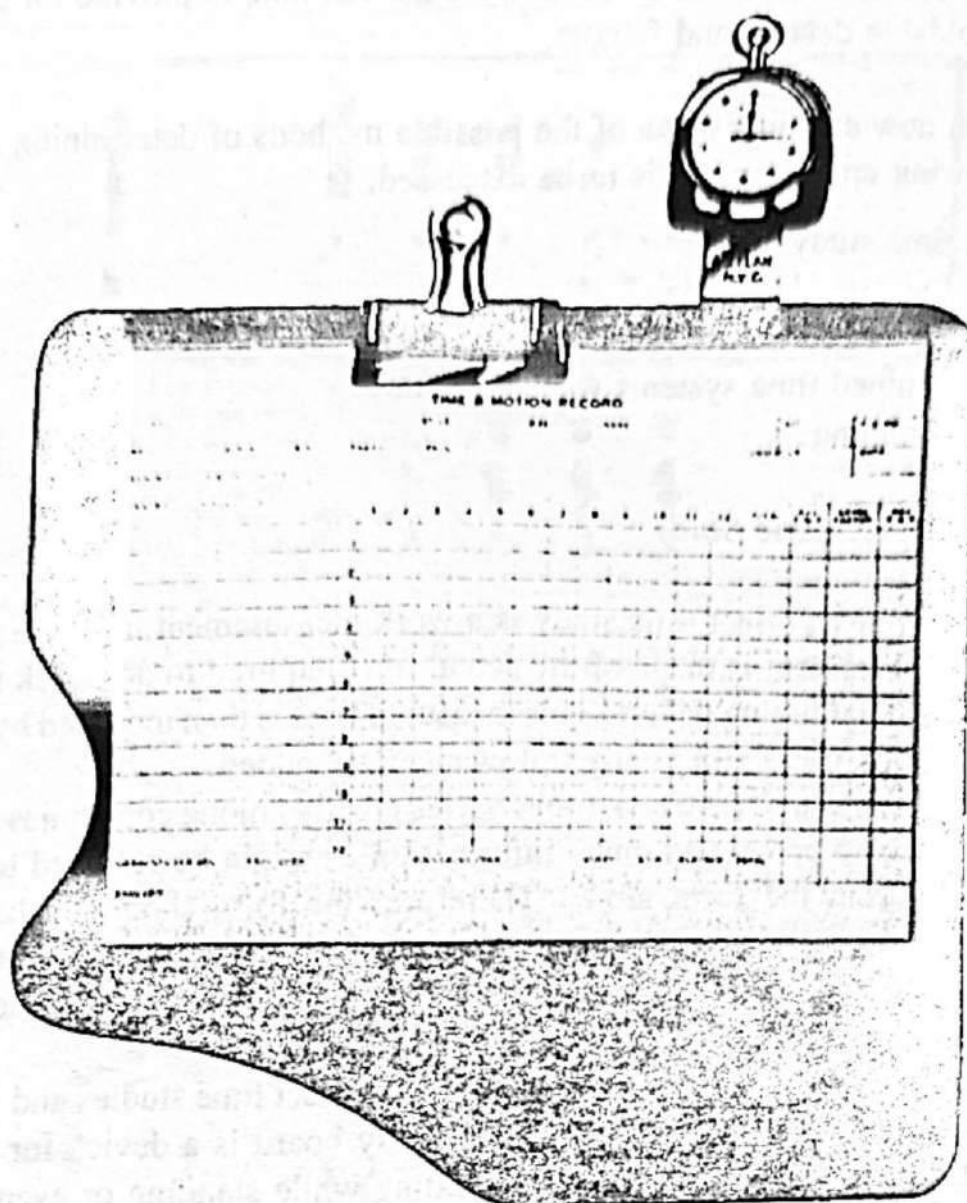


Fig. 6.12. Time study board.

reading time study (the watch is not stopped). The front of the sheet is for recording the task being studied, the existing procedure in detail, and any other pertinent information. The back of the sheet is for recording the actual time study. The task is broken down into easily discernible elements, the elements are recorded, and then the timing is done. The watch reading is recorded in column R, and the time for each element is entered in column T later. For example, the following might be sample results:

Element	<u>1</u>		<u>2</u>		<u>3</u>		<u>4</u>	
	R	T	R	T	R	T	R	T
1	5		18		27		40	
2	10		22		32		45	
3	12		23		35		47	

Element	<u>1</u>		<u>2</u>		<u>3</u>		<u>4</u>	
	R	T	R	T	R	T	R	T
1	5	5	18	6	27	4	40	5
2	10	5	22	4	32	5	45	5
3	12	2	23	1	35	3	47	2

[illegible]

(b)

Fig. 6.13. (cont'd)

Next, each element is averaged. Sometimes a performance rating is applied to each element, but in this case, we shall assume that the entire job (not each element) is rated. The calculations for a *different* example would be as follows (the average time for each element is obtained by averaging all the good readings for that element):

Element	Average
1	15
2	7
3	5
4	8

$$\text{Total actual time} = 15 + 7 + 5 + 8 \\ = 35$$

$$\text{Performance rating} = 115\%$$

$$\text{Normal time} = (35)(1.15) = 40.25$$

$$\text{Allowances} = 12\%$$

$$\text{Standard time} = (40.25)(1.12) = 45.08$$

In this case, the analyst believes that the operator is performing at 115% or 15% faster than would normally be expected. Consequently, the normal time should be adjusted to show that *more time is allowed*. To obtain the normal time, the actual time is multiplied by the performance rating. If the operator has been performing at 90%, or 10% less than what normally would be expected, the normal time would be

$$(35)(0.90) = 28.50$$

Now, realizing that there will be other problems, the problems must be "allowed" for. For example, the operator will have personal needs (rest room, water, etc.) and will become fatigued as the day progresses. Finally, there will be unavoidable delays (machine breakdowns, shortage of parts). We make a study and find that personal needs occupy 4% of the workday, fatigue accounts for 5% of the performance, and unavoidable delays occur 3% of the time. The total allowance then is

$$4\% + 5\% + 3\% = 12\%$$

To calculate the standard time, we add the allowances to the normal time and find

$$\text{Standard time} = (40.25)(1.12) = 45.08$$

The form and example above are highly simplified. No foreign elements (unnecessary moves, for example) were encountered, and performance rating was on the entire job, not on each element. Consequently, real-world time studies can often be very complicated. A competent time study analyst is a highly trained technical person who, since he or she is dealing with other employees almost all the time, must also be a human relations expert.

6.3.2 Time Study Standard Data

Often, elements of a task are repeated many times throughout an organization. Instead of a detailed study of each time a task is performed, the task could be studied

in depth once and a standard data file developed. Then, whenever that element reappears, the time could be directly applied from the standard data file. *Time study standard data* may be defined as normal time values not obtained from direct time measurement of the *particular* element but obtained from direct time measurement of the element in a similar operation earlier.

For example, in a machine shop that produces many small parts all of which are the same approximate size, the element "obtain two parts and place in fixture, X in. to parts and Y in. to fixture" would probably be repeated very often. A left-hand-right-hand chart of the element is shown in Figure 6.14.

After they have been studied, the elements could be placed in a standard data file, such as that shown in Table 6.1. (The values in this table are fictitious.) Then, any time a time study analyst encounters this element, he or she can obtain the normal time directly without having to record stopwatch readings or performance ratings.

The advantages in using standard data are numerous. It must be realized, however, that standard data are not always applicable and that even when they are, some other measurement must be used for unusual elements. A list of advantages given below.

- (1) Individual time studies are quicker and less costly.
- (2) Consistency exists between time studies.
- (3) There is less chance for error on a study.
- (4) Fewer analysts may be needed.
- (5) Some work simplification is done automatically as the method is prescribed.
- (6) It is good for cost estimating and production planning before a job is run.
- (7) There is less disturbance in the shop while the time study is being taken.

The chief disadvantage is that there is a high developmental cost to obtain the necessary data. In some organizations the cost can be completely prohibitive. In

Left-hand Activity	Symbol	Symbol	Right-hand Activity
Reach X "	◊	◊	Reach X "
Grasp	○	○	Grasp
Move Y "	◊	◊	Move Y "
Position	○	○	Position
Release	○	○	Release

Fig. 6.14. Left-hand-right-hand chart of the element, "Obtain two parts and place in fixture, X in. to parts and Y in. to fixture" (time in minutes).

Table 6.1. STANDARD DATA FILE FOR "OBTAIN TWO PARTS AND PLACE IN FIXTURE, X IN. TO PARTS AND Y IN. TO FIXTURE"
(Time in minutes)

X	Y			
	6 in.	8 in.	10 in.	12 in.
6 in.	0.025	0.026	0.026	0.027
8 in.	0.026	0.026	0.027	0.027
10 in.	0.027	0.028	0.028	0.029
12 in.	0.027	0.028	0.029	0.030

general, however, whenever standard data are applicable and the developmental cost is not prohibitive, the data should be used.

6.3.3 Predetermined Times

If jobs or tasks are broken down into finer and finer elements, there will be a point reached where all tasks are made up of the same set of elements. At this point, time values can be assigned to these elements, and the total time required to perform a task can be determined by adding the appropriate elements. This approach is called *predetermined times*.

Predetermined time systems are very similar to standard data in that the time values can be read directly from charts. They are not unique in their application, however, and can be used by any company.

There are several systems in use today. Among the better known are (1) work factor and (2) methods time measurement (MTM).

The time values applied to the various elements can be obtained in several ways. One approach is to use highly accurate movie cameras and use the films (slowed down) to obtain the times (micromotion analysis). Another approach is to use electronic timing devices. Finally, time studies can be used when appropriate.

As an example, one such element might be "move object to other hand," and the time might be recorded as

Weight > 2 lb

<i>Distance</i>	<i>Time</i>
4 in.	0.00412 min
8 in.	0.00591 min

Of course, since the data must be extremely accurate, more decimal places are necessary.

6.3.4 Predetermined Time Standard Data

Predetermined time standard data are standard data exactly as in time study standard data *except* for the source of the information. Here, the repeated elements are timed by using predetermined time systems. For example, Figure 6.14 and Table 6.1 would appear exactly the same, except that now times would be applied to the subelements in the chart of Figure 6.14 from predetermined time systems. Often this distinction is made by saying predetermined systems time elements at a level 1. The level 2 elements (standard data) may be made up of several level 1 elements.

The advantages and disadvantages are exactly the same as those for time study standard data. The only possible difference is that predetermined time standard data may be developed more quickly since all times are already available.

6.3.5 Work Sampling

Work sampling is the estimating of the proportion of time devoted to a given type of activity by the relative frequency occurrence of intermittent randomly spaced instantaneous observations. Suppose, for example, that it is desired to estimate the amount of time that should be set aside for personal needs during a workday or that it is desired to know the amount of time taken up by unavoidable delays such as material shortages or machine breakdowns. This can be done by sampling over an extended period of time. Basically, a series of observations is made randomly (probably through the use of random number tables such as those available in almost all texts on work sampling), and the results are recorded. Sample random numbers are shown below with possible interpretations.

<i>Random Number</i>	<i>Interpretation (When to Observe)</i>	
915	5:15 P.M.	(starting with 100 as 8:00 A.M. and skipping lunch)
723	3:23 P.M.	
019	8:19 A.M.	

Once a series of numbers has been generated, the observations can be made and the percent of time occupied by one activity can be estimated as follows:

$$\text{Percent occurrence for activity A} = \frac{\text{number times A occurred}}{\text{total number of observations}}$$

The number of observations necessary can be determined scientifically, but in general, enough observations are taken so that the accumulated data stabilize. For example, Figure 6.15 shows the typical response and the approximate total number of observations that should be taken— N^* .

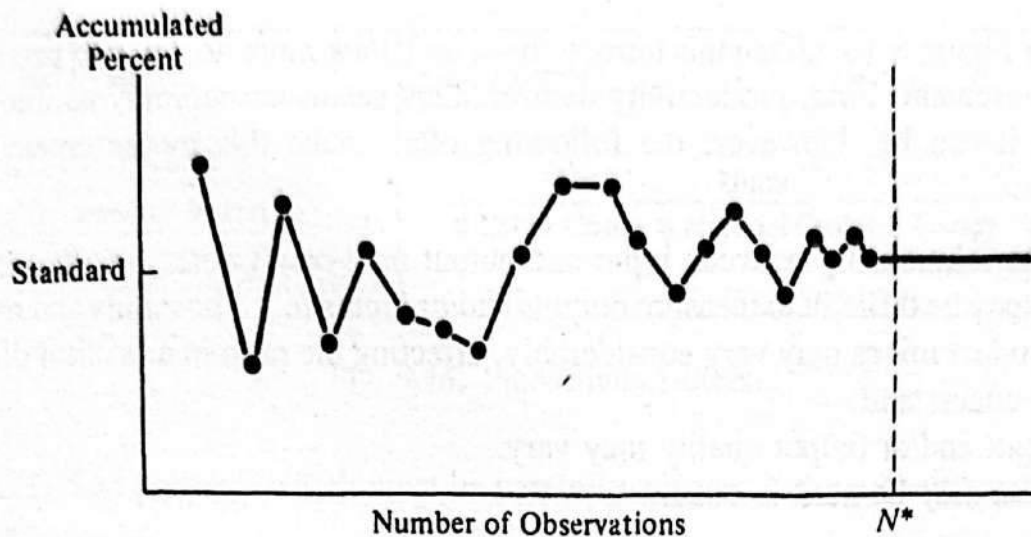


Fig. 6.15. Typical work sampling response.



6.4. Organizational System Performance Measurement

As discussed in Section 6.1, the goal of an organization is usually some composite function of several measures. Profitability, for example, is a composite function bringing together quality, customer satisfaction, sales, cost of labor, material, energy, and others. Naturally, there are other objectives that may not be measured by profitability, but they are usually less important and we will not discuss them here.

A control system designed to optimize profitability is really a series of control subsystems—each with its own subobjectives. For example, a measured day work system controls direct labor by comparing its performance to that determined by standard times (work measurement). This is efficiency measurement in Figure 6.1. Another popular control scheme is quality control.

Recently, the industrial engineering profession has recognized that another control system is needed. Productivity measurement and control is the newest control system and it is receiving a tremendous amount of attention in the literature, at conferences, and in day-to-day discussions. The rest of this chapter will discuss productivity measurement (control is exercised as a result of the measurement system, hopefully).

→ Not new

One must question whether or not productivity measurement is necessary. Are our time-proven systems of quality control and work measurement enough, or do we need more? There is increasing evidence that more is directly needed. Once the leading nation in productivity improvement, we (the United States) are finding our rate of productivity improvement very low—so low, in fact, that we are trailing most industrialized nations in the rate of productivity improvement. Perhaps by measuring and improving productivity, we can reverse this trend.

6.4.1 Productivity Measurement Basics

Productivity measurement is simply a ratio of organizational outputs to organizational inputs [Equation (6.1)]. The ratios can be total or partial. For example,

consider Figure 6.16. Given this introduction, let's look more deeply into productivity measurement. First, productivity measurement seems exceedingly simple, and, in fact, it can be. However, the following often make this measurement more complex:

- (1) The relationship between input and output may be obscure. → Service
- (2) It may be difficult to measure outputs and/or inputs (e.g., hospitals and banks).
- (3) Product mixes may vary considerably, affecting the ratio in a fashion difficult to understand.
- (4) Input and/or output quality may vary.
- (5) Data may be hard to obtain.

In addition to being total or partial measures⁷ (Figure 6.16), productivity indices can be "static" or "dynamic." *Static measures* are ratios taken as a snapshot with no base year comparison. All the ratios in Figure 6.16 are static. *Dynamic measures* are actually ratios of static productivity indices, as shown in the equation

$$\text{dynamic productivity index} = \frac{\frac{\text{outputs this year}}{\text{inputs this year}}}{\frac{\text{outputs base year}}{\text{inputs base year}}} \quad (6.2)$$

Therefore, indices greater than 1 show productivity improvement, while indices less than 1 show productivity deterioration. As with static measures, dynamic measures can be partial or total.

There are two popular productivity measurement approaches. The first uses a group-generated model and is called the *normative productivity measurement methodology* (NPMM). The second is less participative in that one model can be modified to fit any organizational scheme. It is called the *multifactor productivity measurement model* (MFPMM). Each will be discussed in some detail.

6.4.2 Normative Productivity Measurement Model

The NPMM relies on experienced managers to develop the productivity measurement ratios or indices applicable to a specific company. It does this through a structured group process known as the *nominal group technique* (NGT). Although not a panacea to productivity measurement, the NPMM can usually lead to group consensus on what is important to monitor.

The procedure is first to develop the group. Normally, 5 to 12 people are chosen. They should all be familiar with the organization and should represent various interest groups. Give the group a charge. For productivity measurement, the charge might be something like: "Develop a list of productivity measures, ratios, or indices important to our organization." Then the group is immersed into

⁷Sink, "Organizational System Performance."

<u>Partial Measures:</u>	<u>Composite:</u>
Units/Person Hour	Units
Units/Pound of Material	$\$ \text{ Labor Cost} + \$ \text{ Material Cost} + \$ \text{ Energy Cost}$
Units/Million BTU of Energy	



Fig. 6.16. Productivity indices.

the NGT. The process (which must be carefully adhered to by a chairperson) is as follows:

- (1) *Silent Generation*—Each person is asked to generate a list of measures.
- (2) *Round Robin*—After 5 to 15 minutes, the group members are asked to present their ideas. Each person gives a succinct description of one measure. The next person does the same; and so on. This continues around the group as many times as necessary to exhaust everyone's list.
- (3) *Group Clarification*—First, all ideas are clarified. Then overlap or duplication of ideas is identified and remedied. (Normally, several ratios are the same or very similar and can be combined.)
- (4) *Voting*—Each person is given N cards (N is determined by the leader) and asked to list the N best ideas or ratios. After this is done, each person chooses:
 - (a) The best one and gives it N points.
 - (b) The lowest of their list and gives it 1 point.
 - (c) The second best and gives it $N - 1$ points.
 - (d) The second worst and gives it 2 points.
 Etc.

The results might be as shown in Figure 6.17. From these results, we can see two things. First and most important, there is a strong consensus on what is important and what should be monitored. Second, there is some confusion on productivity. Measures 1, 4, and 5 are productivity measures, but 2 and 3 really are not productivity measures (they are more measures of effectiveness than ratios of output to input). This confusion is normal and should be expected.

	Individual Votes (5 = most imp., 1 = least)	Number of Votes/Vote Score (Total)
1. Units per Direct Labor Hour	5-4-4-3-4-5	6/25
2. Number Orders Completed on Time	3-2-2-1-1	5/9
3. Number Customers with Repeat Orders	1-1-3-5	4/10
4. Units per Million BTU (Energy)	4-5-5-4-5	5/23
5. Units per Ton of Purchased Material	2-3-1	3/6

Fig. 6.17. Results from NGT on productivity ratios.

At any rate, there is a strong consensus on units per direct labor hour as being the most important; units per million Btu comes in a close second (perhaps the company is very energy intensive); and so on. From this the list of final measures can be developed and the measurement process started.

Given the list of ranked measures, it is possible to develop a composite model that actually combines these identified ratios through weighting techniques. One such process is known as the *multicriteria performance/productivity measurement technique*.⁸ Details will not be presented here.

6.4.3 Multifactor Productivity Measurement Model

MFPMM is a more comprehensive and analytical approach to measuring changes in productivity. Basically, MFPMM recognizes that there are two ways profitability can be affected in an organization:

- (1) Price recovery.
- (2) Productivity changes.

Under price recovery, prices can be raised or lowered as the market allows or demands. Both can dramatically affect profitability. Price increases can usually be tolerated only as long as competition and consumer reaction eventually occur. The desired change is through productivity improvement(s).

The MFPMM uses accounting techniques to break the total variation (change in profit) into price effects and productivity effects, resulting in a valid productivity measurement model. This analysis can be done on both inputs and outputs. For example, profitability is affected by sales price and sales quantity (outputs). It is also, however, affected by price and quantity changes in labor, energy, materials, and so on (inputs). Careful manipulation of the data will demonstrate what effects each change has.⁹

6.5. Computers and Work Measurement and Design

Computers have had a strong impact on the field of work measurement and design, but particularly on work measurement. For example, several firms use microprocessors directly wired to workstations. This way, a long-range time study can be conducted yielding very accurate measurements.

There has been a great deal of work toward computerizing predetermined time

⁸D. Scott Sink, "Multi-criteria Performance/Productivity Measurement Technique," *Productivity Management*, Vol. II, No. IV, Stillwater, Okla., Fall 1983.

⁹The details of MFPMM techniques are beyond the scope of this chapter and will not be presented. For more, see J. C. Swaim et al., "Current Developments in Firm or Corporate Level Productivity Measurement and Evaluation," *Proceedings 1983 Fall Industrial Engineering Conference*, Toronto, Canada, Nov. 1983.

systems. For example, the computer can take simple coded activities, look up the predetermined times, and do all the calculations. Another more sophisticated approach is to store some previously determined times for higher levels of activities (several predetermined elements as in standard data) and to use these whenever appropriate.

When similar products are produced, it may be possible to store standard times by product descriptions, such as weight or material used. By adequately describing the product to the computer, the appropriate standard time could be obtained. Interpolation could be used for given characteristics that have not previously been calculated.

Finally, there are numerous routines available on computers that generate random numbers. These random numbers can then be used in the design of a work sampling procedure.